Intrusion Detection in Industrial IoT Using Deep Learning

Qawsar Gulzar and Khurram Mustafa

Abstract The phrase "Industrial Internet of Things" (IIoT) refers to a range of devices, such as actuators, databases, services, industrial applications, and human resources, that are present in the workplace. To maintain the stability of IIoT-connected devices, it is crucial to have a strong security framework in place that safeguards user data and thwarts illicit actions. Therefore, deploying an intrusion detection system is crucial to effectively detect and classify various forms of assaults and anomalous activities inside the IIoT environment. This study employs a deep learning system to forecast several cybersecurity threatens, encompassing distributed denial of service (DDoS), brute force, spoofing, denial of service (DoS), reconnaissance, web-based, and Mirai assaults. The study presents a novel and effective recurrent neural network (RNN) model that precisely forecasts these attack categories by utilizing the Pearson correlation coefficient. To assess the performance of the RNN model, it was compared to various models like the decision tree (DT), random forest (RF), XGBoost, and artificial neural network (ANN). The evaluation was based on accuracy, precision, recall, and $F1$ score metrics. The efficiency of the existing intrusion detection system (IDS) model in identifying different attacks and anomalies was evaluated using the CICIoT 2023 dataset. To ensure a fair and impartial representation of the data, the synthetic minority over-sampling method (SMOTE) was employed. The test results indicate that the suggested recurrent neural network (RNN) model attained an accuracy of 99.15%. This was accomplished with a learning rate of 0.01, a dropout rate of 0.5, and a prediction time of 30.51 ms. The precision, recall, and $F1$ scores achieved 99.22%, 99.23%, and 99.23%, respectively. The proposed approach improves the effectiveness of intrusion detection by an average of 4.67% when compared with the most sophisticated machine learning algorithms used for securing Internet of Things (IoT) devices.

Keywords IIoT · RNN · Pearson correlation · Deep learning · Intrusion detection

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1 Introduction

The Industrial Internet of Things (IIoT) is a specialized version of the Internet of Things (IoT) that is specially tailored for use in industrial settings. The IIoT enhances an industry's capacity to operate with increased efficiency and reliability.

An industrial system's surveillance and management functions are greatly enhanced in an intelligent manufacturing system [1] by incorporating diverse cyber-physical systems and contemporary communication methods [2]. To comprehend what Industry 4.0 is, the next wave of industrial transformation, it is essential to grasp the idea of smart manufacturing. The industrial market incorporates many sensors, actuators, and innovative technology. An up-to-date evaluation forecasts that the worldwide market for IoT devices will achieve a valuation of 75.4 billion US dollars by the year 2025. Reliability, reaction time, and network latency are crucial considerations in contemporary business. Given these variables, it is imperative to improve data transfer and decision-making technological advances to operate autonomously, eliminating the need for human intervention. The IoT has emerged as a very captivating area of research in recent years due to its widespread application in connecting an infinite number of consumer products to offer convenience and amenities for consumers' everyday lives [3, 4]. Following the principles of Industry 4.0, incorporating the IoT into industrial settings significantly improves productivity, efficiency, and safety [5]. In essence, the successful integration of IoT into industrial environments is intricately linked to the IIoT. We can most accurately describe the IIoT as a four-tiered architecture [6]. Within the industrial context, this architectural framework has many levels, including the "physical," "network," "middleware," and "application" layers, as seen in Fig. 1. The "physical layer" refers to a range of physical infrastructure, including gadgets, sensors, mobile and computer devices, and other monitoring and automation equipment that is now being used [6]. The "network layer", includes various communication networks such as "wireless sensor networks", "cellular networks", and "machine-to-machine interfaces."

The "middleware layer" features web services, application programming interfaces, and cloud storage, facilitating interaction between the "network and application layers." At the pinnacle of IIoT architecture, the "application layer" facilitates many *industrial functionalities and services*, encompassing smart buildings, healthcare, transportation, robotics, and beyond.

The IIoT is a comprehensive framework that caters to many sectors and industries. However, technology introduces numerous new challenges related to security, privacy, legal concerns, and social interactions. To tackle these problems, it is necessary to implement solutions that may be easily expanded and adapted. IoT devices have limited resources and need security solutions capable of meeting limited storage requirements, low power consumption, and affordable pricing. These solutions need to be in line with accepted communication norms. IIoT systems are very susceptible to attacks because of the large volume of data produced by IoT devices during industrial operations [7, 8]. Conventional data processing methods are inadequate for managing the vast amounts of data generated by IoT- and IIoT-based applications.

Fig. 1 Industrial IoT architecture

Machine learning (ML) is often regarded as a highly suitable computing approach for incorporating intelligent capabilities into IoT devices.

According to published research, intrusion detection systems have been developed using statistical models, rule-based models, artificial intelligence methods, and classical machine learning and data mining techniques [9]. These approaches often result in high false positive rates (FPRs) because of the overlap between genuine and anomalous data. This study introduces an innovative multiclass lightweight recurrent neural network (RNN) prediction model that leverages Pearson correlation to predict specific attack classes and mitigate high FPRs. This model automatically analyzes raw data to identify abnormal patterns efficiently. Unlike shallow networks like artificial neural networks (ANNs), which typically have only one or two hidden layers, deep networks consist of multiple layers with diverse architectural styles. So deep learning (DL) successfully handles extensive dimensional data and extracts the underlying structure from the labelled data.

Our research contributions are as follows:

- The creation and deployment of a specialized IDS utilizing deep learning techniques, specifically engineered to classify multiple types of attacks in IIoT networks.
- A performance comparison and evaluation between ML and DL classifiers.

- An assessment is conducted to determine the efficacy of the suggested methodology by generating multiple performance metrics under varying constraints, using the CICIoT 2023 dataset, a recent dataset related to security.

The following sections of the article are organized in the following manner: Sect. 2 provides a succinct summary of current studies on security on the IoT and IIoT. Section 3 explores the process of preparing the dataset, selecting relevant features, and designing the overall system architecture. Section 4 offers a lucid elucidation of the mathematical underpinnings of the proposed model. Section 5 provides a comprehensive explanation of the process for setting up the physical components and computer programs, as well as an examination of the gathered outcomes. The last portion functions as the conclusion of the research investigation.

2 Previous Work

Effectively overseeing and regulating extensive industrial systems in the IIoT poses complex and severe problems. Computing systems must efficiently and securely handle and analyze large amounts of industrial data within a certain timeframe [10, 11]. In addition, for low latency and reliable data transfer, *the system's capacity and throughput should be high*. The use of ML techniques and models has greatly enhanced the reliability and safety of the industrial sector. The security issues in IIoT systems might be greatly reduced using these methods [12, 13]. The following is current research that examines machine learning-based security strategies for both the IoT and the IIoT.

An unsupervised method of feature learning is used by the authors in [14] in their non-symmetric deep autoencoder (NDAE). The authors constructed their proposed model utilizing TensorFlow, a software framework that incorporates graphics processing unit (GPU) functionalities. The research utilized the NSL-KDD dataset, achieving an attack detection rate of 89.22%. In [15] researchers successfully discovered and classified hazards related to the Internet of Things (IoT) using support vector machines (SVM) and Bayesian approaches. Their model was applied to the "KDD CUP 99 dataset" and achieved an accuracy score of 91.50%.

In their study, the researchers in [16] devised a method that integrates wavelet transform and deep neural networks to detect fraudulent data injection assaults. The researchers utilized the IEEE 118 dataset to implement their proposed method. Their suggested model had a 91.80% accuracy in identifying assaults. The authors in [17] devised a technique to detect intrusions by detecting departures from typical patterns. Their methodology effectively identified denial of service (DoS), Man-in-the-Middle (MitM), and Structured Query Language (SQL) injection vulnerabilities in IIoT and IoT applications. The researchers assessed the effectiveness of their suggested technique by utilizing the NSL-KDD dataset, and their model achieved a 91.65% accuracy in identifying assaults.

In their study, the researchers in [18] presented a technique for detecting intrusions in IoT systems. This approach makes use of a sophisticated learning algorithm that is based on linear discriminant analysis. The authors assessed the precision of the proposed methodology by employing the NSL-KDD dataset. Their method attained a level of accuracy of 92.35%.

The authors in [19] utilized deep learning (DL) and transfer learning techniques to bolster the security of Internet of Things (IoT) networks, utilizing embedding layers to enhance the performance of the BoT-IoT dataset through a personalized deep learning approach.

In their study, the researchers in [20] to achieve optimum detection recognition, they combined the "*spider monkey optimization (SMO)*" method with the "*stacked-deep polynomial network (SDPN).*"

In [21] the researchers provided a unique method for data preprocessing used in a lightweight intrusion detection system (IDS), using both ML and DL classifiers. The BoT-IoT and TONIoT network datasets were utilized for the tests and evaluation. In their study, the authors of [22] devised an efficient intrusion detection system intending to protect IoT networks from DDoS attacks. Their overall accuracy percentage reached an amazing 90%.

Overall, most researchers have put out threat detection systems that focus on applications within IoT and IIoT. Their models were mainly assessed using NSL-KDD and KDD Cup 99 open-source datasets. These datasets were used extensively and are designed for IIoT applications. Hence, it is essential to have new datasets to adhere to the current security requirements of IIoT networks. A further constraint of the pioneering research is that most researchers have failed to address the adaptability of their suggested models to devices with constrained resources. This paper aims to resolve these obstacles by implementing a novel deep learning-based method for identifying attacks. This approach utilizes a recently developed dataset focusing on IIoT devices' security. This study introduces a unique and efficient neural network approach to anticipate attacks in IIoT networks. This model utilizes the CICIoT 2023 dataset and other datasets, one of the most recent and comprehensive datasets available.

3 Proposed Method

This article proposes IIoT threat detection using the simple recurrent neural network (RNN) model. The attack detection infrastructure comprises a confluence of several operations. Figure 2 illustrates the technique for detecting attacks. The collection and analysis of datasets is the first stage of this design. At this juncture, the dataset was assembled and examined based on the type of information included. Subsequently, the dataset underwent preprocessing, including tasks such as data cleansing, feature engineering, and standardization. All these methods were used to acquire the data features. The 80% and 20% ratio, respectively, was used to partition these vectors of features into a training dataset and a testing dataset. The suggested recurrent neural

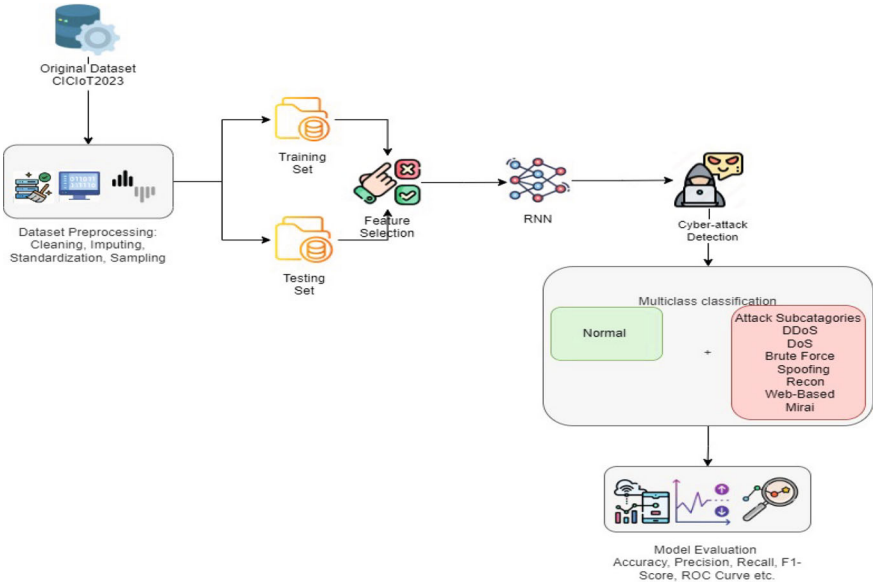

Fig. 2 Flowchart for the intrusion detection in IIoT

network was trained on the training set to facilitate learning. The ensuing model was assessed using a test set based on certain evaluation measures.

3.1 Dataset Description

To assess the exactness of our proposed recurrent neural network (RNN) model, we used the CICIoT 2023 dataset benchmark, the latest iteration provided by the Canadian Institute for Cyber Security (CIC). In this analysis, we examine seven distinct types of anomalies: "DDoS, brute force, spoofing, DoS, recon, web-based, and Mirai." These categories are outlined in Table 1.

The dataset includes a multitude of cyberattacks as outlined in Table 1. It comprises 447,775 records employed for training and 89,555 records employed for testing. There are 47 characteristics involved. In the suggested methodology, the dataset undergoes an initial preparation phase where extraneous data is eliminated, and any missing items are substituted. The most prominent features are then determined by employing the select k-best technique from the sklearn.feature_selection package, after the Pearson correlation coefficient technique was applied. We implemented Python-TensorFlow on a central processing unit (CPU) equipped with an Intel i7 core to simulate our proposed model. Regarding the process of selecting features, we seeded the RNN model with 32 features and conducted 100 epochs in total.

Table 1 Elaborate depiction of the attacks included inside the CICIoT 2023 dataset

Attack category	Description	Examples
DDoS	Those attacks include overloading IoT operations with threats from multiple sources to disrupt access to them	*ACK fragmentation, UDP flood, SlowLoris, ICMP flood, RSTFIN flood, PSHACK flood, HTTP flood, UDP fragmentation, ICMP fragmentation, TCP flood, SYN flood, SynonymousIP flood* [23]
DoS	Like DDoS attacks, DoS attacks usually originate from a single source	*TCP flood, HTTP flood, SYN flood, and UDP flood* [23]
Recon	The adversary is attempting to collect all available information to prepare for future actions. One may carry out these assaults in a variety of ways	Ping sweep, OS scan, vulnerability scan, port scan, and host discovery
Web-based	Numerous methods are used by web-based attacks to target online services	Injections, browser hijacking, backdoor malware, uploading attack, cross-site scripting (XSS) and DoS
Brute force	To gain illegal access to systems, brute-force attacks include repeatedly trying different data combinations, such as passwords or passphrases. "*The objective of an attack is to systematically attempt each word provided in the dictionary to ascertain the right password*"	Dictionary brute-force
Spoofing	Fraudulent users may "take on the identity of a target system to intercept network communications without authorization" using spoofing attacks. This technique aims to gain illegal access to computer systems, steal data, and spread dangerous software	ARP spoofing, DNS spoofing
Mirai	The Mirai assault is a widespread DDoS that specifically targets IoT devices. The assault included the infiltration of devices to create a botnet capable of overwhelming specific targets via flooding	GREIP, GREETH, and UDP plain

3.2 Dataset Preprocessing

The primary objective of the succeeding step is to ascertain the nature of the features. The collection mostly comprises numerical and category data.

3.2.1 Handling Missing Values

Data preprocessing often involves many steps, including the primary and first phase of addressing the missing values. Many statistical parameters, such as mean and standard deviation, i.e. imputing, are utilized in different data preprocessing procedures to substitute missing values or occurrences. This dataset is provided cleanly, meaning it has no missing entries.

3.2.2 Handling Categorical Data

Several attributes in the dataset include a wide range of string parameters, which are inappropriate for most machine learning tasks. Python has many libraries for one-hot encoding (OHE). This may be achieved using the one-hot encoder from the Sklearn preprocessing library. These techniques help deal with categorical data by converting text values into numeric values based on the data classifications in a specific feature. One of the robust Python packages for performing one hot encoding of categorical data is called Scikit-learn. The suggested model utilizes the OHE approach from the Sklearn preprocessing library.

3.2.3 Data Standardization

A proper normalizing of the values after one-hot encoding is necessary. A standard scaler is used in the proposed model to standardize the input data and place all the numerical characteristics on the same scale.

3.2.4 Avoiding Class Imbalance Problems

To overcome class imbalance difficulties, as shown in Table 3, we used the most popular synthetic minority over-sampling approach (SMOTE) to generate synthetic cases of the minority classes. This is beneficial in achieving dataset balance and enhancing classifier performance, mainly when the minority class is marginalized.

3.3 Feature Selection

To evaluate the proposed model's efficacy without running into overfitting issues, we started the investigation by using a significant choice of the most favourable set of features from the input dataset, which is of utmost importance. However, opting for the most features in a conventional machine learning classifier increases the likelihood of overfitting. The Pearson correlation coefficient, denoted as 'r', quantifies a linear relationship's strength and trajectory involving a pair of continuous

variables. The correlation coefficient spans from − 1 to + 1. An absolute negative correlation is shown by a value of − 1, whereas a perfect positive correlation is represented by a value of 1, and a value of 0 indicates no correlation. A number of − 1 represents an absolute negative correlation, whereas a value of 1 shows a perfect positive correlation. A value of 0 signifies no correlation. The Pearson correlation coefficient may be calculated using the following mathematical formula:

$$r = \frac{\sum \left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)}{\sqrt{\sum \left(X_i - \overline{X}\right)^2 \left(Y_i - \overline{Y}\right)^2}}, \tag{1}$$

where X_i and Y_i refer to data points, whereas $\overline{X}$ and $\overline{Y}$ correspond to the means of the X and Y variables, respectively.

Positive r values suggest a direct link, implying that as one element increases, the other also tends to rise. In contrast, negative r values suggest a negative correlation, indicating that when one variable grows, the other tends to decrease. A correlation is considered better as the value of r approaches 1 or − 1. Conversely, r values below 0 indicate a weak or non-existent linear association.

The feature covariance matrix, shown in Fig. 3, displays the prominent attributes of the CICIoT 2023 dataset. It reveals that some pairings of features exhibit a very high correlation of 99.99%. We eliminated 15 features {'AVG', 'LLC', 'Magnitude', 'Max', 'Number', 'Radius', 'Srate', 'Std', 'Totsize', 'Weight', 'ack_count', 'ack_flag_number', 'fin_count', 'rst_flag_number', 'syn_count'} from the figure because their correlation was not favourable as per the threshold value of Pearson correlation coefficient (i.e. the value of 0.8).

4 Mathematical Description Behind the Proposed Method

4.1 Data Modelling

The data modelling procedure begins after splitting the data frame. For a classification model, the data frame is split into two parts: 80% for training and 20% for testing. The proposed model employs the "train_test_split" function from the "sklearn.model_selection" library, with a testing proportion of 20% and a random state of 42.

This work suggests a recurrent neural network method in conjunction with Pearson correlation as a feature selection strategy to foster the classification accuracy of cyberattack detection in IIoT. The overall method of cyberattack detection based on the suggested recurrent neural network is shown in Fig. 1. Training and test datasets have been generated for model verification.

After receiving data from the X sensor of dimension d at time t, the suggested model's outcomes are categorized as benign or malicious. After sufficient training, the models may be used to categorize new data samples.

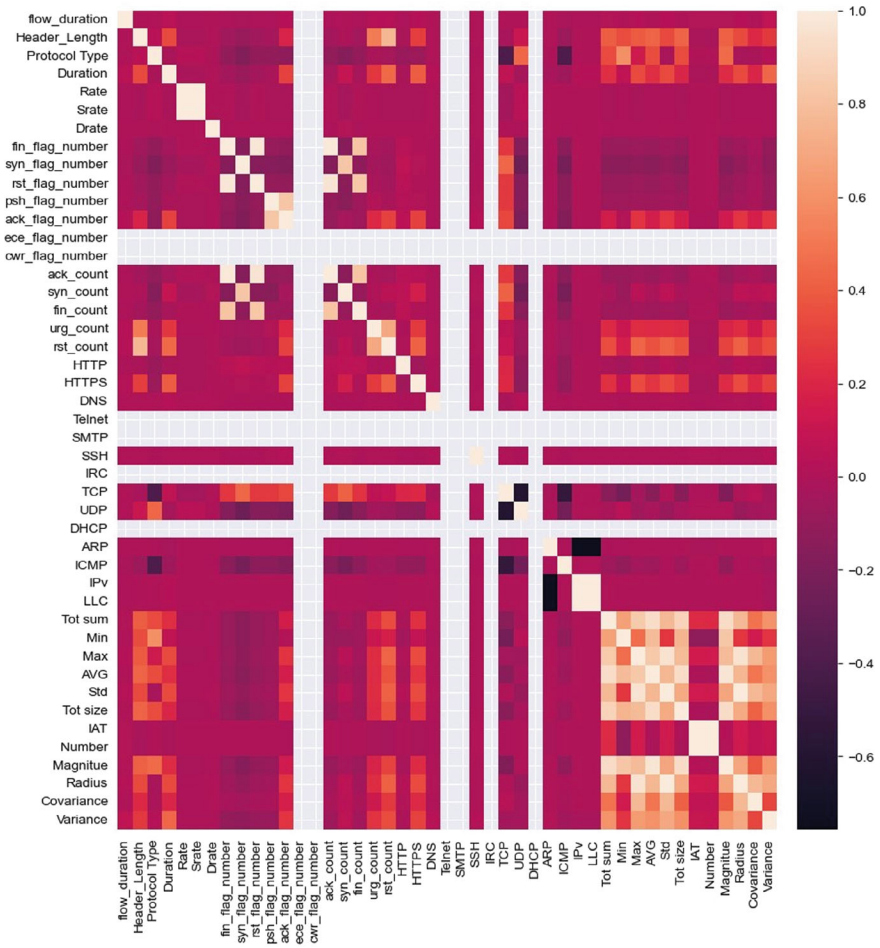

Fig. 3 Correlation analysis of the attributes in the CICIoT 2023 dataset. A correlation coefficient 1 indicates an intense positive association, whereas a correlation coefficient of − 0.6 indicates an intense negative correlation. The presence of empty areas in certain features indicates that the Pearson correlation was impossible to compute. This is because the value of that feature stayed constant throughout the whole dataset

The simulation design of the suggested model for detecting cyberattacks in the IIoT is shown in Table 2. It includes the RNN model. We employed an RNN model to enhance the precision and diminish the computational intricacy in the identification of cyberattacks on IIoTs. Out of these models, the RNN model exhibited superior performance compared to the most advanced models in terms of results as well as exhibited lesser training times. RNNs exhibit a high degree of suitability for analyzing and predicting IIoT data. Its ability to capture patterns and relationships makes it valuable for detecting anomalies and forecasting tasks. To overcome the

nonlinearity issue, the explored deep neural network uses the CICIoT 2023 dataset and feature selection to find appropriate features.

The RNN model consists of two layers. The first layer, a simple RNN layer, consists of 128 units, and the last layer, a dense layer, consists of 1 unit. Between the two layers, we applied the dropout regularization rate of 0.5 and the learning rate from 0.1 to 0.001 to allow the model to overcome the overfitting and underfitting problems, respectively.

The following mathematical equations determine the RNN model's outputs y_t^{RNN} and hidden layers h_t^{RNN} for a given feature vector $\{x_1^{\mathrm{RNN}}, x_2^{\mathrm{RNN}}, \ldots x_t^{\mathrm{RNN}}\}$:

$$h_t^{\mathrm{RNN}} = g_h\left(w_i x_t^{\mathrm{RNN}} + w_R h_{t-1}^{\mathrm{RNN}} + b_h\right), \tag{2}$$

$$y_t^{\mathrm{RNN}} = g_y\left(w_y h_t^{\mathrm{RNN}} + b_y\right). \tag{3}$$

The weighted matrices w_i, w_R, w_y represent the connections between a neural network's input, output, and hidden layers. The bias vectors b_h and b_y are additional parameters that are used to adjust the outcome of the network. The variable t denotes the count of samples, while r represents the chosen activation function. The activation functions that may be considered are the rectified linear unit (ReLU) and Softmax.

Table 2 Suggested approach's criteria for training

Parameter	Value(s)
Language	Python
Libraries	Pandas, Numpy, Matplotlib, Sckitlearn, Matplotlib, Keras, and Scipy
Train set	80%
Test set	20%
RNN	[1]
Neurons per RNN layers	[128, 1]
Dropout	[0.5]
Dense layers	[1]
Neurons per dense layer	[128, 1]
Activation function	[ReLU, Softmax]
Optimizer	Adam
Learning rate	0.1–0.001
Epochs	100
Batch size	64

Table 3 Table displays the performance measures outcomes and confusion matrix values for each analyzed attack

Attack category	Attack subcategory	Accuracy	Precision	Recall	$F1$-score	TPR	TNR	FPR	FNR
DDoS	DDoSACK_Fragmentation	0.9998	0.9942	1.0	0.9971	1.0	0.9998	0.0001	0.0
	DDoS-UDP_Flood	0.9801	0.6658	0.6684	0.6671	0.6684	0.9897	0.0102	0.3315
	DDoS-SlowLoris	0.9993	0.9800	0.9986	0.9892	0.9986	0.9993	0.0006	0.0013
	DDoS-ICMP_Flood	0.9994	0.9844	0.9978	0.9910	0.9978	0.9995	0.0004	0.0021
	DDoS-RSTFINFlood	0.9996	0.9946	0.9917	0.9931	0.9917	0.9998	0.0001	0.0082
	DDoS-PSHACK_Flood	0.9993	0.9855	0.9930	0.9892	0.9930	0.9995	0.0004	0.0069
	DDoS-HTTP_Flood	0.9999	0.9991	0.9981	0.9986	0.9981	0.9999	0.0000	0.0018
	DDoS-UDP_Fragmentation	0.9994	0.9914	0.9894	0.9904	0.9894	0.9997	0.0002	0.0105
	DDoS-ICMP_Fragmentation	0.9999	0.9997	0.9990	0.9993	0.9990	0.9999	0.0	0.0009
	DDoS-TCP_Flood	0.9999	0.9999	0.9996	0.9997	0.9996	0.9999	0.0	0.0003
	DDoS-SYN_Flood	0.9786	0.5809	0.9153	0.7108	0.9153	0.9799	0.0200	0.0844
	DDoS-SynonymousIP_Flood	0.9991	0.9759	0.9963	0.9860	0.9963	0.9992	0.0007	0.0036
Brute force	DictionaryBruteForce	0.9910	0.9780	0.7113	0.236	0.7113	0.9995	0.0004	0.2886
Spoofing	DNS_Spoofing	0.9841	0.6619	0.9516	0.7808	0.9516	0.1006	0.0015	0.0483
	MITM-ArpSpoofing	0.9830	0.6532	0.8771	0.7488	0.8771	0.9857	0.0142	0.1228
DoS	DoS-SYN_Flood	0.9995	0.9913	0.9919	0.9916	0.9919	0.9997	0.0002	0.0442
	DoS-UDP_Flood	0.9834	0.7306	0.7094	0.7198	0.7094	0.9919	0.0080	0.2905
	DoS-TCP_Flood	0.9980	0.9509	0.9835	0.9669	0.9835	0.9984	0.0015	0.0164
	DoS-HTTP_Flood	0.9989	0.9850	0.9776	0.9813	0.9776	0.9995	0.0004	0.0223
Recon	Recon-PingSweep	0.9840	0.8338	0.5725	0.6788	0.5725	0.9965	0.0034	0.4274
	Recon-OSScan	0.9841	0.9133	0.5095	0.6541	0.5095	0.9985	0.0014	0.4904
	VulnerabilityScan	0.9823	0.8077	0.5262	0.6372	0.5262	0.9962	0.0037	0.4737

(continued)

Table 3 (continued)

Attack category	Attack subcategory	Accuracy	Precision	Recall	$F1$-score	TPR	TNR	FPR	FNR
	Recon-PortScan	0.9846	0.7564	0.6892	0.7212	0.6892	0.9934	0.0065	0.3107
	Recon-HostDiscovery	0.9946	0.8912	0.9305	0.9105	0.9305	0.9966	0.0033	0.0694
Web-based	SQL injection	0.9944	0.9264	0.8816	0.9034	0.8816	0.9978	0.0021	0.1183
	CommandInjection	0.9998	0.9976	0.9956	0.9966	0.9956	0.9999	0.0	0.0043
	Backdoor_Malware	0.9917	0.8483	0.8726	0.8603	0.8726	0.9953	0.0046	0.1273
	Uploading_Attack	0.9819	0.7084	0.6550	0.6806	0.6550	0.9918	0.0081	0.3449
	XSS	0.9997	0.9924	1.0	0.9961	1.0	0.9997	0.0002	0.0
	BrowserHijacking	0.9858	0.7458	0.7771	0.7611	0.7771	0.9921	0.0078	0.2228
Mirai	Mirai-greeth_flood	0.9995	0.9865	0.9983	0.9924	0.9983	0.9995	0.0004	0.0016
	Mirai-greip_flood	0.9999	0.9983	1.0	0.9991	1.0	0.9999	0.0	0.0
	Mirai-udpplain	0.9990	0.9750	0.9928	0.9838	0.9928	0.9992	0.0007	0.0071
Benign	–	0.9996	0.9888	0.9995	0.9941	0.9995	0.9996	0.0003	0.0004
	Total	0.8868	0.9162	0.8868	0.8851	0.8868	0.9410	0.0033	0.1286

A combination of benign and malicious network traffic data from the corresponding attack was used to train, validate, and test the model for each outcome

4.2 Evaluation Metrics

We tested the usefulness of our projected model using established performance criteria, including accuracy, precision, recall, $F1$-score, and other metrics, within the context of research on spotting cyberattacks in IIoT environments.

1. Accuracy: The classification accuracy is calculated by dividing the number of accurately classified cases by the total number of occurrences:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{4}$$

2. Precision (PPV): The "Positive Predictive Value (PPV)" or precision is a quantitative metric that assesses the proportion of accurate positive classifications out of the overall number of positive classifications:

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{5}$$

3. Recall (DR)/Sensitivity (TPR): The classifier's "*True Positive Rate (TPR)*":

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{6}$$

4. Specificity: The classifiers "*True Negative Rate (TNR)*":

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \tag{7}$$

5. False Positive Rate (FPR): A ratio of mistakenly anticipated positive outcomes out of the total actual negative outcomes.

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{8}$$

6. False Negative Rate (FNR): A ratio of mistakenly anticipated negative outcomes out of the total actual positive outcomes.

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \tag{9}$$

7. $F1$-Score: Denotes a state of perfect balance between "precision" and "recall", where a score of 1 represents flawless accuracy and recall:

$$F1\text{-Score} = \frac{2 \times \text{PPV} \times \text{TPR}}{\text{PPV} + \text{TPR}} \tag{10}$$

8. Receiver Operating Characteristics (ROC): Receiver operating characteristic (ROC) curves are used in machine learning to evaluate binary classification models. This curve shows the model's sensitivity (true positive rate or recall) and specificity (true negative rate) trade-off across categorization thresholds.

$$\mathrm{ROC} = \frac{\mathrm{TPR}}{\mathrm{FPR}} \tag{11}$$

The AUC-ROC measures the overall performance of a model that categorizes data into two groups. The ROC curve is quantified by computing the area under the curve, a numerical value that spans from 0 to 1.

5 Results and Discussion

5.1 Software Implementation

The RNN model is implemented and evaluated on a Microsoft Windows 11 Pro device with an Intel(R) Core (TM) i7-10610U CPU @ 1.80 GHz, running at a frequency of 2304 MHz. It is then compared to several machine learning classifiers. The device had 4 cores and 8 logical processors, 32 GB RAM, and a 512 GB SSD. The model has been evaluated using the Python 3.11.4 (64-bit) edition, employing numerous libraries like Numpy, Pandas, Matplotlib, Keras, Scipy, Tensorflow, and SkLearn.

5.2 Anomaly Detection and Attack Classification by a Classifier

A classification report showing whether the input is a part of the regular traffic heading into an IoT network or if it is part of a malicious/attack category is the output sent by the classifier. All experiments utilize a labelling technique called "one-hot encoding" conducted to represent the attack classes. In addition, the classifier generates confusion metrics, accuracy %, and Region of Convergence (ROC) curve to assess the results.

Several deep learning methods and machine learning approaches published in the literature are applied to the CICIoT 2023 dataset. A confusion matrix, *error matrix*, or *contingency table* is a means that may be utilized to evaluate and predict the effectiveness of a classification model. Figure 4a shows the binary class confusion matrix, and Fig. 4b shows the multiclass confusion matrix for 33 attack types in the proposed dataset. Furthermore, Table 3 showcases the performance measures

computed on the testing set, in conjunction with the outcomes of the confusion matrix. The formulae shown in Sect. 4.2 serve as the foundation for the performance measures.

Table 3 displays the observed balance between malicious (abnormal) and regular requests, together with the low proportion of "*FP's*", and "*FN's*" estimated by the RNN model throughout the various tests. The suggested technique's effectiveness

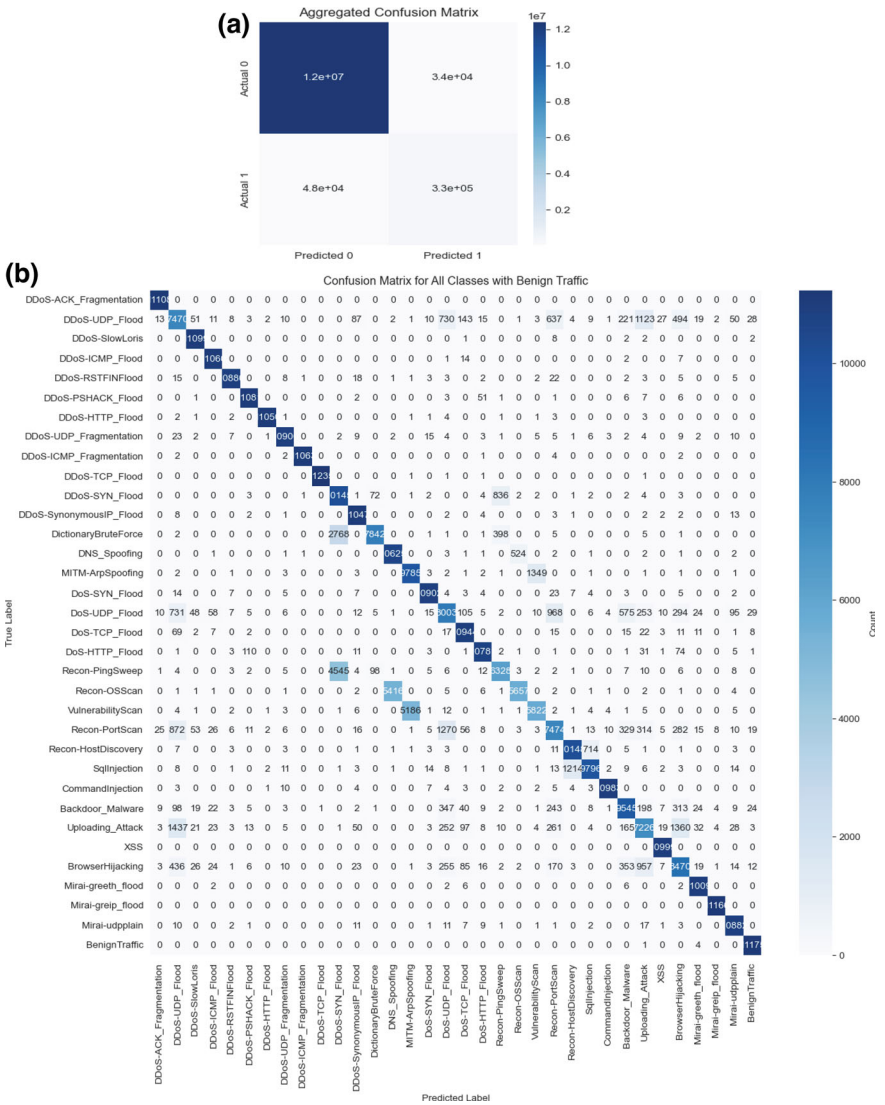

Fig. 4 **a** Binary class confusion metrics of CICIoT 2023 dataset; **b** multiclass confusion metrics of CICIoT 2023 dataset based on the test set validation

for the CICIoT 2023 dataset and all the attacks in the proposed dataset are shown by the ROC curves in Fig. 5a, b. The area beneath the diagonal lines shows the model's fairness and the AUC values for each attack class in the CICIoT 2023 dataset. All three images demonstrate that the multicolour curves are positioned close to the peak in the top left corner, indicating near-optimal or flawless performance in accurately identifying both positive and negative observations. In addition, the RNN model demonstrates reliability and robustness by regularly achieving high performance on the CICIoT 2023 dataset shown in Fig. 5c.

5.3 Evaluation Against the State-of-the-Art Methods

To evaluate the general efficacy and dependability of the suggested recurrent neural network (RNN) model, its performance was contrasted with four alternative classifiers: XGBoost, random forest (RF), decision tree (DT), and artificial neural network (ANN). The XGBoost classifier demonstrated worse accuracy in comparison with the other classifiers, obtaining training and test accuracies of 88.91% and 86.55%, respectively. The decision tree (DT) algorithm fared better than XGBoost, with a training accuracy of 88.96% and a testing accuracy of 86.78%. The RF classifier exhibited greater performance in comparison with XGBoost and DT, with a training accuracy of 89.75% and a testing accuracy of 88.33%. The ANN surpassed all other classifiers, with a training accuracy of 99.28% and a testing accuracy of 99.02%. Nevertheless, the recurrent neural network (RNN) outperformed all other classifiers, producing the most favourable outcomes (Table 4).

Table 5 provides a succinct summary of the performance comparison between the proposed RNN and the latest available security techniques. This study examined the most recent anticipated designs from 2018 to 2023. Multiple researchers employed the "NSL-KDD and KDD-CUP 99 datasets" to investigate their research questions and assessed the efficacy of their models by assessing the accuracy of attack detection. We have selected the CICIoT 2023 dataset, which is a current dataset specifically related to IIoT security. This dataset includes numerical data that encompasses all occurrences of attacks, making it a valuable resource for comparison with other benchmark datasets. The proposed recurrent neural network (RNN) has shown higher accuracy of 99.15% for the test dataset highlighted in bold in detecting attacks compared to current sophisticated machine learning (ML) classifiers.

6 Conclusion

This paper presents a novel and efficient method that utilizes a lightweight recurrent neural network (RNN) to identify specific threats and abnormalities in Industrial Internet of Things (IoT) devices. The study investigated a range of attack methods, such as distributed denial of service (DDoS), brute-force assaults, spoofing, denial

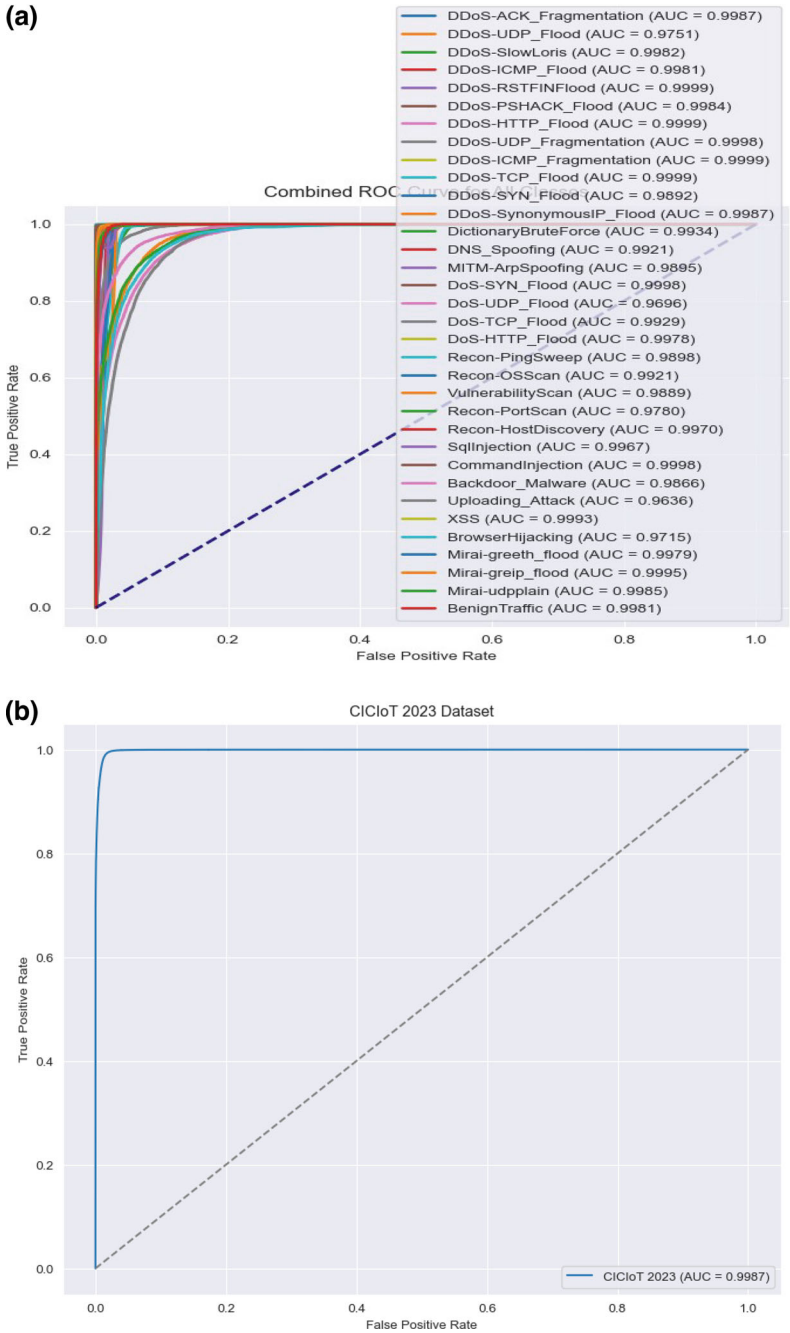

Fig. 5 **a** ROC curve for the CICIoT 2023 dataset's multi-attack classes; **b** ROC curve for the CICIoT 2023 dataset

Table 4 Comparative analysis of XGBoost, RF, DT, ANN, and RNN performance

Performance parameters		XGBoost	Algorithms			
			Machine learning		Deep learning	
			RF	DT	ANN	RNN
Training	Accuracy	88.91	89.75	88.96	99.28	99.35
	Precision	89.00	92.00	90.00	99.26	99.34
	Recall	81.00	89.00	88.00	99.28	99.35
	$F1$-score	83.00	90.00	88.00	99.27	99.35
Testing	Accuracy	86.55	88.33	86.78	99.02	99.15
	Precision	88.54	91.83	89.21	99.00	99.22
	Recall	80.04	88.19	87.88	99.02	99.23
	$F1$-score	82.99	89.34	87.42	99.01	99.23

Table 5 Analysis comparing the RNN model with the most advanced methods

Reference	Dataset	Intrusion detection technique	Accuracy (%)
[14]	NSL-KDD	Non-symmetric deep autoencoder	89.22
[15]	KDD CUP'99	Hybrid GA-SVM with PSO	91.50
[16]	IEEE 118	Deep neural networks	91.80
[17]	NSL-KDD	Anomaly-based IDS	91.65
[18]	NSL-KDD	LDA-based extreme learning machine (ELM)	92.35
[21]	BoT-IoT	Customized deep learning technique	–
[20]	NSL-KDD	Stacked deep polynomial network (SDPN)	92.66
[22]	BoT-IoT, and TON_IoT	ANN, LSTM, NB, LSVM	99.00
Our work	CICIoT 2023	Recurrent neural network	**99.15**

of service (DoS), reconnaissance, web-based attacks, and Mirai deployment. The proposed recurrent neural network (RNN) method outperforms existing approaches in successfully detecting these assaults, reaching an accuracy rate above 99.15% with a processing time of 30.51 ms, as specified in Table 4. The recurrent neural network (RNN) achieved its highest level of performance when trained with a learning rate of 0.01 and a dropout rate of 0.5. When comparing with other machine learning approaches such as XGBoost, decision trees (DT), random forest (RF), and artificial neural networks (ANN), the proposed RNN model showed superior accuracy and higher values for additional metrics such as recall and $F1$ score. The primary advantage of our model is its ability to identify all attack categories expected in the CICIoT 2023 dataset, as well as in other industrial IoT environments with similar network traffic patterns.

Nevertheless, further comprehensive experiments may be carried out based on the outlined future trajectory to improve the suggested model's accuracy and applicability. Future research will also be done in more depth and real-time on the other neural network models and datasets.

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Code The code is accessible from the corresponding author.

Availability of Data and Material This paper provides all the data in figures and tables.

Competing Interest No competing interest as reported by the scholar.

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